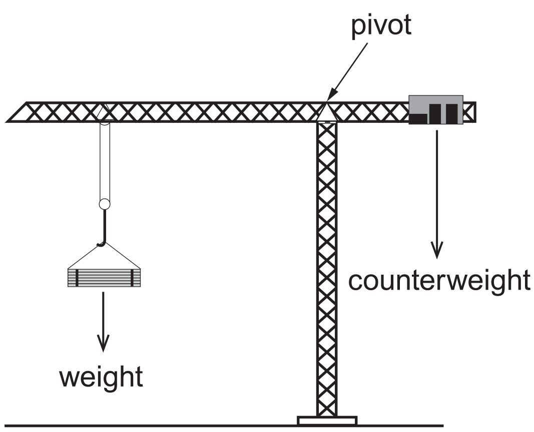
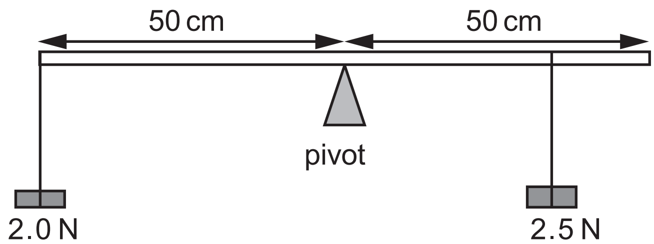
Examiner only

5. The diagram shows a tower crane used on a new building in the centre of Cardiff. A weight is lifted on the long arm and a counterweight on the short arm keeps the crane from toppling over.



(a) A group of students investigate a model of the crane by hanging weights from a horizontal metre ruler.



They hang a 2.0 N weight at one end, 50 cm from the pivot and another weight of 2.5 N somewhere on the other side of the pivot to make it balance.

(i) Use the equation:

moment = force × distance

to calculate the anticlockwise moment (in N cm) of the 2 N weight about the pivot. [2]

Moment = ... N cm

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(ii) Use your answer to part (i) and the equation:

$$\text{distance} = \frac{\text{moment}}{\text{force}}$$

to calculate the distance that the 2.5 N weight should be placed from the pivot for the ruler to balance. [2]

Distance = ... cm

(iii) Explain why it would be impossible to balance the 2 N weight on the end of the ruler if the 2.5 N weight is replaced with a 1.5 N weight. [2]

....

....

....

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(b) A particular tower crane is capable of lifting a maximum mass of 15 000 kg.
Use the equation:

$$\text{weight} = \text{mass (kg)} \times \text{gravitational field strength}$$

to calculate the weight of this mass. [2]
[gravitational field strength, g = 10 N/kg]

Weight = ... N

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